

SULIT
DP024
Physics 2
Semester II
Session 2022/2023
2 hours

DP024
Fizik 2
Semester II
Sesi 2022/2023
2 jam



KOLEJ MATRIKULASI KEDAH

PRA PEPERIKSAAN SEMESTER PROGRAM MATRIKULASI (PRA PSPM)

JANGAN BUKA KERTAS SEHINGGA DIBERITAHU.

DO NOT OPEN THIS QUESTION PAPER UNTIL YOU ARE TOLD TO DO SO

Kertas soalan ini mengandungi ___ halaman bercetak.

This question paper consists of ___ printed pages.

Answer **all** questions.

1 (a)

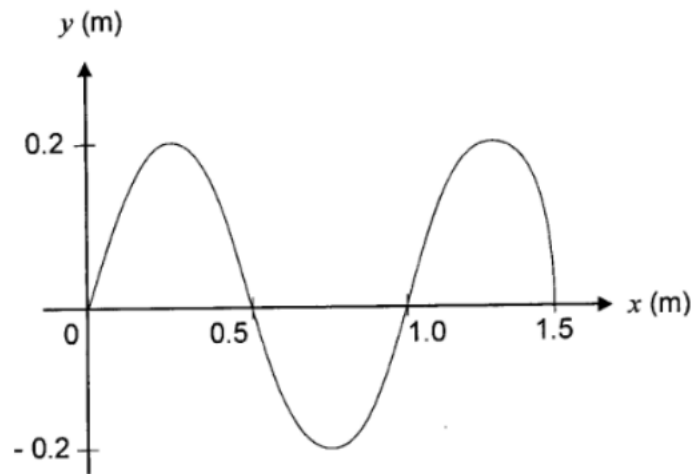


FIGURE 1

FIGURE 1 shows the displacement-distance graph of longitudinal wave travelling with a speed of 5 m s^{-1} in the positive x -direction. Determine the

- (i) wave equation that represents this wave.
- (ii) particle vibration velocity at $t = 0 \text{ s}$, $x = 0.20 \text{ m}$.

[6 marks]

(b) Displacement equation of stationary wave is given by,
$$y = 3 \cos(0.1\pi x) \sin(100\pi t)$$
where y and x in unit cm. Determine the

- (i) amplitude of particle when $x = 2 \text{ cm}$
- (ii) angular frequency
- (iii) wave number
- (iv) speed of wave

[7 marks]

(c) A stretched wire of length 3 m and mass 920 g vibrates transversely in a string with tension 15 N . Calculate the

- (i) mass per unit length of the wire.
- (ii) wave speed in a stretched string.

[3 marks]

- 2 (a) Calculate the number of electrons that flow in a wire if it carries a current of 2 A for 5 s.

[2 marks]

- (b) A conducting wire has a 1.0 mm diameter, a 2.0 m length and a 50 mΩ resistance. Calculate its resistivity.

[3 marks]

- (c)

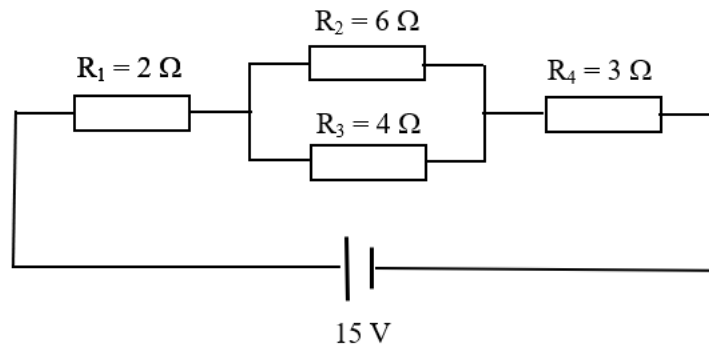


FIGURE 2

FIGURE 2 shows $R_1 = 2\ \Omega$, $R_2 = 6\ \Omega$, $R_3 = 4\ \Omega$ and $R_4 = 3\ \Omega$ are connected to a 15 V power supply. Calculate the

- (i) effective resistance.
(ii) voltage across 4 Ω resistor.

[7 marks]

3 (a)

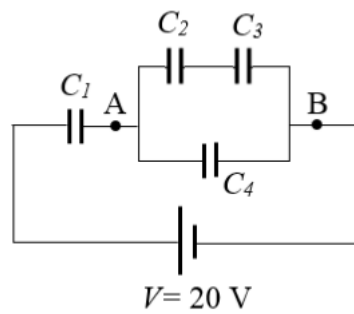


FIGURE 3.1

FIGURE 3.1 shows a combination of four capacitors. All the capacitors are identical and each has a capacitance of $30 \mu\text{F}$. The potential difference between point A and B is 8.0 V . Calculate the

- (i) effective capacitance between A and B.
- (ii) the potential difference across capacitor C_1

[5 marks]

(b)

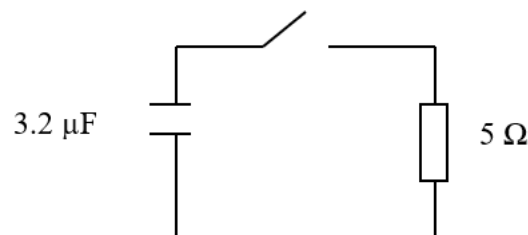


FIGURE 3.2

FIGURE 3.2 shows a fully charged $3.2 \mu\text{F}$ capacitor, connected to a 5Ω resistor and a switch. The potential difference across the capacitor is 15.0 V .

- (i) Calculate the charge stored in the capacitor.
- (ii) Calculate the time constant, τ of the circuit.
- (iii) Sketch a graph of charge against time after the switch is closed. Indicate the time constant, τ on the graph.

[5 marks]

- 4 (a)
(i)



FIGURE 4.1

FIGURE 4.1 shows the top view of two straight current-carrying conductor. Sketch the magnetic field lines produced by the two conductors.

[2 marks]

- (ii) Calculate the magnetic field strength at the center of solenoid with $l = 30 \text{ cm}$, number of turns $N = 15$. The current flows in the solenoid is 2.5 A .

[2 marks]

- (b)

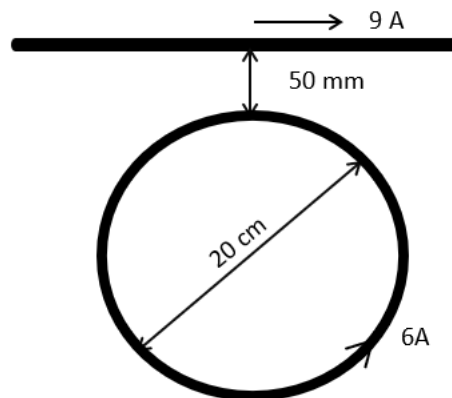


Figure 4.2

Figure 4.2 shows a straight wire parallel to a circular single loop. The diameter of the loop is 20 cm and the wire is 50 mm from the loop. The currents in the wire and the loop are 9 A and 6 A , respectively. Determine the magnitude and direction of the resultant of magnetic field at the center of the loop.

[5 marks]

- (c) A coil of 100 turns and cross sectional area 0.2 m^2 is lying in a magnetic field in which the magnetic flux changes from 2.0 mT to 7.5 mT in 120 ms . Calculate the

- (i) change in magnetic flux experienced by the coil
- (ii) e.m.f. induced in the coil
- (iii) current that flows in the coil, if the resistance of the coil is 10Ω .

[5 marks]

- 5 (a) An object is placed 15 cm from a curved mirror. An image which is twice the size of the object is formed in front of mirror.

- (i) Is the mirror convex or concave? Explain your answer.
- (ii) Determine the radius of curvature of the mirror.

[6 marks]

- (b) An orange is placed 25.2 cm in front of a diverging lens with a focal length of 18 cm.

- (i) Sketch the ray diagram to show the formation of the image.
- (ii) Determine the image distance.
- (iii) Determine the magnification.
- (iv) State two (2) characteristics of the image.

[8 marks]

- 6 (a) Two narrow slit separated 2.4 mm are illuminated by a light with wavelength 512 nm. The screen is placed 6.5 m from the slits. Determine the distance between

- (i) adjacent bright fringes on a screen.
- (ii) the fifth dark fringe and seventh bright fringe.

[8 marks]

- (b) A 475 nm light passes through two narrow slits. The interferences pattern is observed on a screen at a distance 85.0 cm from the slits. The second order bright fringe is seen at ± 2.01 cm from the central bright fringe. Calculate the

- (i) slit separation.
- (ii) width of second dark fringe.

[6 marks]